Domed Monuments Cluster at Babylonian Bēru Harmonics: A Longitude Enrichment Test on the UNESCO World Heritage List

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Abstract

A one-dimensional angular enrichment test is applied to the UNESCO World Heritage Cultural and Mixed corpus ($N = 1011$) using the 0.1-bēru interval (3°), the finest integer subdivision of the Babylonian unit (MUL.APIN, c. 1000 BCE). Mount Gerizim (35.269°E) serves as anchor on historical grounds (§6.1). Sites within ≤ 0.002 bēru (∼6.7 km) of a harmonic node are Tier-A+. Domed monuments ($N = 90$) show significant A+ enrichment (Bonferroni $p_{\text{adj}} = 0.003$, $k = 3$), and cluster asymmetry independently reaches significance ($p_{\text{adj}} = 0.01$). The anchor is historically motivated; the pipeline is fully reproducible.

Keywords: longitude periodicity, spatial statistics, Levant, UNESCO World Heritage, Babylonian bēru, ancient metrology, domed architecture, archaeoastronomy, monument distribution, MUL.APIN

1 Introduction

This study applies a one-dimensional angular enrichment test to the externally curated UNESCO Cultural and Mixed corpus ($N = 1011$), using a reproducible computational pipeline (see Data and Code Availability, §A.2). The approach employs a geometric null with no free parameters and is intended as a reusable framework for spatial pattern-detection studies in archaeological heritage data.

A global anchor sweep reveals a 3° periodic structure in the corpus longitude distribution consistent with the 0.1-beru unit; the full treatment and its methodological implications are in §5.3 and §6.1.

Why this design addresses key methodological criticisms. Prior metrological investigations were rightly criticized for investigator-assembled site lists (Williamson & Bellamy, 1983; Freeman, 1976; Kendall, 1974). Three features distinguish this study: (1) the UNESCO site list was compiled without reference to this hypothesis and is fixed;

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(2) the 0.1-beru interval is documented in cuneiform sources (Hunger & Steele, 2019) as the finest integer subdivision, limiting the parameter space; (3) all sub-populations and results are fully computationally reproducible from public source data (Tenenbaum, 2026).

Positioning within quantitative archaeometry. This paper belongs to the tradition of probability-based spatial analysis in archaeometry (Magli, 2013, 2016; Ruggles, 1999, 2015; Silva, 2020).

2 Background

2.1 The Beru Unit

The *beru* (30° of arc) is attested in the MUL.APIN cuneiform compendium (Hunger & Steele, 2019; Watson & Horowitz, 2011), dated to the late second or early first millennium BCE. The unit divided the ecliptic into twelve segments and served for both stellar and terrestrial angular measurement; the terrestrial use is explicit in MUL.APIN II ii 43–iii 15. The sub-division into 30 *uš* (1 uš = 1°) is well attested (Hunger & Steele, 2019; Robson, 2008), and the 0.1-beru interval (3°) tested here corresponds to exactly 3 *uš*.

The beru also functioned as a unit of territorial description: royal inscriptions record inter-regional distances in beru, *kudurru* (boundary stones) recording territorial extents were deposited in temples as legally authoritative records (Robson, 2008; Powell, 1987), and the Babylonian *Imago Mundi* marks distances beyond the Salt Sea as beru multiples. Systematic meridian transit observation and eclipse timing, both attested in the cuneiform record (Hunger & Steele, 2019; Neugebauer, 1975), are sufficient to derive inter-site angular separations at 1° resolution; the gnomon, attested across Mesopotamian, Egyptian, and Indian traditions (Neugebauer, 1975; Shamasastry, 1915), provides an independent means of establishing true meridian. This combination provides the documentary basis for the hypothesis; the causal question of precisely how or why this unit is applied is outside scope.

2.2 Ancient Metrology and Geodetic Practice

Whether pre-modern societies employed standardized angular measures across large geographic scales has a long and contested history. Thom's "megalithic yard" (Thom, 1967, 1971, 1964) stimulated statistical refinement (Kendall, 1974) and methodological critique (Freeman, 1976). Powell's treatment of Mesopotamian metrology (Powell, 1987) remains the standard reference. More recently, Magli (Magli, 2013, 2016) has applied Monte Carlo methods to archaeoastronomical alignments, and Ruggles and Silva have advanced probability-based frameworks for evaluating orientation claims (Ruggles, 1999, 2015; Silva, 2020). The ICOMOS-IAU thematic study (Ruggles & Cotte, 2010) provides institutional context for archaeoastronomical analysis of World Heritage properties. Medieval cartographic traditions also employed directional reference systems anchored to sacred geography, with Jerusalem and Mecca serving as orienting meridians in mappamundi and qibla charts (Harvey, 1987), providing historical context for the Levantine corridor hypothesis.

This study departs from prior work in three respects: the site list is externally curated (UNESCO World Heritage), eliminating selection bias; the test is one-dimensional (longitude only), yielding a well-defined geometric null (4%) with no free parameters; and the analysis is fully computational and publicly archived (Tenenbaum, 2026).

2.3 The Anchor: Mount Gerizim

Mount Gerizim (35.269°E) is the primary reference anchor because of its documented *tabbur ha-aretz* ("navel of the earth") role in Samaritan cosmographic tradition (Purvis, 1968; MacDonald, 1964; Crown, 1989), analogous to Babylon's meridian in cuneiform astronomical records (Hunger & Steele, 2019). As a prominent mountain (881 m) in the central highlands, it would have served as a natural survey reference point. No optimization across alternative Levantine anchors was performed.

Gerizim's tentative-list coordinate (ref. 5706, DMS E35 16 08) is externally sourced and does not appear in the UNESCO inscribed corpus (UNESCO World Heritage Centre, Tentative List ref. 5706, State of Palestine, submitted 02 April 2012; archived at `https://whc.unesco.org/en/tentativelists/5706/`). Jerusalem (35.2317°E), approximately 4.1 km east of Gerizim, is an inscribed corpus member. It is self-excluded when used as anchor and produces the same A+ count (§5.3). Both lie within the *Levantine corridor* (∼35°E, Jordan Valley to Levant), and the enrichment belongs to this corridor, not to the specific anchor within it. Mecca reaches only $p \approx 0.057$ (ns); Mt. Meru (Tanzania, 36.750°E) reaches $p \approx 0.305$ (ns) and Mt. Kailash, the most cited geographic candidate for the cosmological axis mundi, reaches $p \approx 0.206$ (ns). Neither alternative anchor approaches significance.

3 Data and Methods

3.1 Corpus Construction

The UNESCO World Heritage XML dataset contains 1248 inscribed nominations across all categories. Two outcome-blind filters yield the analysis corpus: (1) Natural-only sites are excluded (−235, retaining Cultural and Mixed); (2) entries lacking geocoordinates are excluded (−2). The result is $N = 1011$ sites with valid longitudes.

3.2 Geodetic Framework

The reference anchor is Mount Gerizim at 35.269°E, from Palestine's UNESCO tentative list (ref. 5706, 2012; see §2.3). Gerizim is not on the inscribed World Heritage List and therefore does not appear in the standard corpus. Jerusalem is an inscribed corpus member, scored as a regular site in Tests 1–4 (A+ at 4.1 km under Gerizim). For the anchor-comparison audit and both sensitivity sweeps (§5.3), Gerizim is added as a synthetic corpus entry so that each anchor can self-exclude symmetrically: Gerizim removes itself and tests against the $N = 1011$ inscribed sites (including Jerusalem); Jerusalem removes itself and tests against the same N (now including Gerizim).

For a site at longitude λ, the beru-distance is $B = |\lambda - 35.269|/30$ and the beru-deviation from the nearest 0.1-beru harmonic is $\delta = |B - \mathrm{round}(B, 0.1)|$. The tier classification is a fixed angular criterion. Kilometer labels (e.g. $\lesssim 6.7$ km for Tier-A+) are equatorial approximations computed as $\delta \times 30 \times 111$ km/°, becoming smaller at higher latitudes. This makes the angular criterion *conservative* at high latitudes. A geodesic sensitivity check confirms that applying a strict physical threshold would increase the A+ count to 70, strengthening all primary results. No latitude correction is applied.

The geometric null for Tier-A+ is $2 \times 0.002/0.1 = 4\%$, assuming uniform longitudes. This serves as a convenient reference, but the primary tests are simulation-based null

models (§5.5) that preserve the empirical longitude distribution. A Gaussian KDE produces 40.4 ± 6.2 expected A+ ($\sim$4.0%), confirming the geometric null is not materially inflated by longitude clumping.

3.3 Metrological Basis for the 0.1-Beru Spacing

The harmonic spacing tested is 0.1 beru (3°), the smallest integer sub-division of the beru attested in cuneiform metrology, equalling exactly 3 *uš* (Hunger & Steele, 2019; Robson, 2008; Powell, 1987). It was selected because it is the finest metrologically attested integer sub-division that yields a tractable test (120 harmonics, 4% geometric null). Coarser spacings (0.5 beru, 1 beru) are uninformative; finer sub-divisions (0.02 beru) do not produce the signal (Table 6). No spacing was selected post-hoc.

The 0.05-beru spacing (6°) also reaches significance (Table 6). This is structurally expected because every 0.10-beru harmonic is also a 0.05-beru harmonic, making the 0.05-beru result a coarser sub-sampling of the same periodicity, not a separate signal.

3.4 Tier Classification

Sites are classified by δ, the angular deviation from the nearest 0.1-beru harmonic. All tier boundaries were fixed on metrological grounds, not tuned to the data. **Tier A++**: $\delta \leq 0.0002$ beru ($\lesssim$ 0.67 km), $P_0 = 0.4\%$, descriptive only. **Tier A+**: $\delta \leq 0.002$ beru ($\lesssim$ 6.7 km), $P_0 = 4\%$ (i.e. $2 \times 0.002/0.1$), primary threshold. **Tier A**: $\delta \leq 0.010$ beru ($\lesssim$ 33 km), $P_0 = 20\%$. **Tier B**: middle zone — neither near a harmonic nor near a midpoint. The C-band is the symmetric mirror of the A-band, measured as $\delta_{\text{mid}} = 0.050 - \delta$ (distance from the nearest inter-harmonic midpoint rather than from the nearest harmonic): **Tier C**: $\delta_{\text{mid}} \leq 0.010$ beru ($\lesssim$ 33 km), $P_0 = 20\%$. **Tier C−**: $\delta_{\text{mid}} \leq 0.002$ beru ($\lesssim$ 6.7 km), $P_0 = 4\%$. **Tier C−−**: $\delta_{\text{mid}} \leq 0.0002$ beru ($\lesssim$ 0.67 km), $P_0 = 0.4\%$, descriptive only. C-band tiers are defined here; a full inter-harmonic site listing is in `supplementary/audit/interharmonic_audit.txt`.

3.5 Test Populations

Keyword populations are approximate; the raw sweep tolerates known false positives while the context-validated variant (Test 2x) applies sentence-level gating for greater precision. Both populations are reported.

Test 1 (full corpus, primary). Full corpus ($N = 1011$), no keyword filter; binomial test of whether the global A+ rate exceeds the 4% geometric null.

Test 2 (dome/spherical, primary). 7 morphological keywords from four roots (*stupa/stupas*, *tholos*, *dome/domed/domes*, *spherical*) are matched as whole words against each site's name, XML description, and extended OUV text ($N = 90$). This raw sweep is over-inclusive: it admits sites where the keyword refers to geological or vernacular features rather than monumental architecture. The context-validated population ($N = 83$) is reported alongside all primary results; it yields stronger significance, consistent with the rejected sites being non-architectural false positives.

Test 2b (mound evolution, sensitivity variant). Extends Test 2 by adding earthen mound-stage keywords (*tumulus, tumuli, barrow, barrows, kofun*). Not counted separately in the Bonferroni family.

Table 1: Multiple-comparisons summary. Primary test family: Tests 1–3 ($k = 3$, $\alpha = 0.05/3$).

Test	N	Raw p	p_{adj}, k=3	Status
Test 1 (full corpus)	1011	0.0103^a	0.031 *	**Primary**
Test 2 (dome A+)	90	0.0077^a (OR = 2.71)	0.003 ***	**Primary**
Test 3 (cluster asym.)	1011	0.0033^b	0.01 **	**Primary**
Test 4 (temporal)	1011	0.0481^b	—	Descriptive
Test 2b (mound evol.)	110	0.0049^a (OR = 2.67)	N/A	Sensitivity
Test 5 (founding)	1011	0.009^a	N/A	Post-hoc

[a]Fisher exact p vs. rest of corpus (one-sided, enrichment); [b]permutation p. Binomial baselines: Test 2 $p = 0.0009$ ***, Test 2b $p = 0.0005$ ***.

Test 3 (cluster asymmetry, primary). Full corpus ($N = 1011$), no keyword filter; tests whether A+ harmonics host more total sites than non-A+ harmonics, independent of any morphological keyword.

Test 4 (temporal gradient, descriptive). Cochran-Armitage trend test on the full corpus based on year of inscription into the UNESCO corpus, no keyword filter.

Test 5 (founding classifier, post-hoc). A context-aware keyword classifier (`lib/founding_filter.py`) refined after initial data inspection to reduce false positives in population construction, with residual interpretive ambiguity and some false positives accepted (Appendix A.2). It addresses what kinds of sites beyond domed monuments tend to fall on harmonic nodes, but carries no weight in the primary evidential chain.

3.6 Multiple-Comparisons Protocol

Primary test family ($k = 3$).

- **Test 1** (full corpus, $N = 1011$): global enrichment above 4% null.
- **Test 2** (domed/spherical monuments, $N = 90$): morphological sub-population enriched above 4% null.
- **Test 3** (cluster asymmetry, full corpus, no keyword filter): A+-bearing harmonics host more total sites.

Tests 2 and 3 address structurally different questions on different populations and carry the primary evidential weight; Test 1 provides the corpus-level backdrop within the same family. All three pass the Bonferroni bound at $k = 3$ (full corpus $p_{\text{adj}} = 0.031$ *, dome $p_{\text{adj}} = 0.003$ ***, cluster $p_{\text{adj}} = 0.01$ **).

Other tests. Test 4 (temporal gradient, Cochran-Armitage) is descriptive context; the trend is substantially explained by regional composition (§4.5) and is not part of the primary family. Test 2b is a sensitivity variant of Test 2. Test 5 (founding classifier) is post-hoc (Appendix A.2). Test B (Rayleigh, §4.4) is a structural description of the A+ phase-lock that is tautological by construction and is not included in either table.

Results of all primary and secondary tests are consolidated in Table 1.

A comprehensive FDR audit (44 tests; Benjamini & Hochberg 1995) retains 27 at $q < 0.05$, of which 6 survive full Bonferroni at $k = 44$. All three primary results (Tests 1–3) pass FDR correction; Tests 2 and 3 additionally survive full-k Bonferroni. The 6 full-k survivors are: domed/spherical A+ enrichment (Test 2, $p_{\text{adj}} = 0.0416$); harmonic-level Mann-Whitney ($p < 0.0001$); sequential-peak permutation ($p = 0.00025926$); corridor occupancy 17/17 harmonics with binomial $p = 1.7 \times 10^{-24}$ and Fisher $p = 1.5 \times 10^{-19}$;

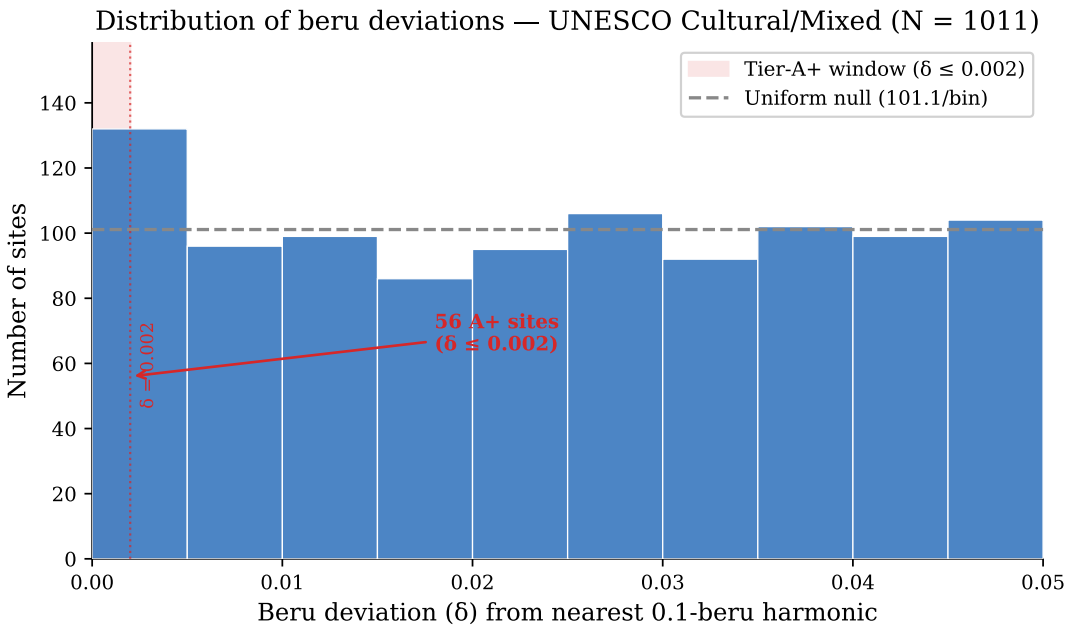

Figure 1: Distribution of beru deviations for the full UNESCO corpus. The excess of sites within the Tier-A+ window ($\delta \leq 0.002$ beru, shaded) above the uniform null (dashed line) establishes the corpus-level signal against which Tests 2 and 3 demonstrate morphological and spatial specificity.

and evolution combined A+ (Fisher $p = 0.0049$ **, binomial $p = 0.0005$ ***). The full audit is in the archived repository (Tenenbaum, 2026).

3.7 Null Models

Three simulation approaches (code: `simulation_null_model.py`, 100,000 trials each) and two dedicated geographic-concentration nulls are described with their results in §5.5.

4 Results

Subsections follow test numbering (Test 1, 2, 2b, 3, 4, 5).

4.1 Global Enrichment (Test 1, Primary)

The full UNESCO Cultural/Mixed corpus ($N = 1011$) shows a Tier-A+ rate of 5.5% (Figure 1) (56/1011; Clopper-Pearson 95% CI [4.2%, 7.1%]), above the 4% null (binomial $p = 0.0103$; Bonferroni $p_{\text{adj}} = 0.031$, $k = 3$). The signal belongs to the ∼35°E corridor, not to either anchor individually (§5.3). The full-corpus enrichment ratio (1.38×) is modest compared to the dome (3.06×) and cluster asymmetry results, consistent with only a fraction of sites driving the periodicity.

Region-stratified robustness. A region-specific empirical null test (Fisher combination of per-region binomials) yields $p = 0.8230$ (ns; pre-2000: $p = 0.4205$, ns): when each region's own A+ rate is the null, the stratified signal does not reach significance. The pre-2000 cohort remains significant against the geometric null ($p = 0.0041$ **); see Test 4 (§4.5) for the full region-adjusted analysis.

World religion sites

Religion sub-populations are defined by keyword sets in `keywords.json` (`religion_sets`); each set extends beyond the bare religion name to include architecturally and historically associated terms (e.g. *cathedral*, *crusader*, *byzantine*, *biblical* for Christianity; *stupa*, *bodhi*, *ashoka* for Buddhism; *synagogue*, *hebrew*, *israelite*, *biblical*). The full per-religion keyword lists and all matched sites are tabulated in `supplementary/audit/religion_keyword_audit.txt`. Sites associated with any world religion ($N = 577$): 6.1% A+ (1.52×; Fisher exact vs. rest: OR = 1.27, $p = 0.2414$, ns; binomial $p = 0.0108$, *). Christianity ($N = 397$): 6.8% A+ (OR = 1.47, $p = 0.1028$, ns; binomial $p = 0.0058$, **), with a marginal A-tier signal (23.4%; OR = 1.09, $p = 0.3228$, ns; binomial $p = 0.0522$, ∼). Islam ($N = 178$): 6.7% A+ (OR = 1.30, $p = 0.2695$, ns; binomial $p = 0.0552$, ∼). Buddhism ($N = 71$): 5.6% A+ (OR = 1.02, $p = 0.5651$, ns) taken alone, but Buddhist sites show a bimodal distribution across the harmonic spectrum: A-tier (20/71, 28.2%) and C-band (19/71, 26.8%) are both elevated simultaneously; the joint probability is $p = 0.0052$ ** (multivariate hypergeometric, equivalent to a three-group Fisher exact). Judaism ($N = 50$): both Tier-A+ (5/50, 10.0%, binomial $p = 0.0490$) and Tier-A (16/50, 32.0%, binomial $p = 0.0308$) nominally exceed the geometric baseline, but neither survives Fisher exact comparison against the rest-of-corpus rate (A+: OR = 1.98, $p = 0.1370$; A: OR = 1.66, $p = 0.0749$), likely reflecting the modest sample size ($N = 50$).

4.2 Domed and Spherical Monuments (Test 2, Primary)

The keyword filter is described in §3.5 ($N = 90$). Tier-A+ enrichment: 11/90 sites (12.2% [6.3, 20.8], 3.06×; Fisher OR = 2.71, $p = 0.0077$ **; binomial $p = 0.0009$ ***; permutation $Z = 3.21$, $p = 0.004$; bootstrap $p = 0.006$; Bonferroni $p_{\text{adj}} = 0.003$, $k = 3$). Tier-A enrichment: 29/90 (32.2% [22.8, 42.9], 1.61×; Fisher OR = 1.72, $p = 0.0175$ *; binomial $p = 0.0043$ **). The χ^2-uniform test (5 bins, df = 4) is not significant ($p = 0.0732$), consistent with the signal being concentrated at harmonic nodes rather than spread uniformly. Tier-B (middle zone): 45/90 sites; mean $\delta = 0.0220$ beru.

Tier-A+ sites ($N = 11$, 12.2%, 3.06×, Bonferroni $p_{\text{adj}} = 0.003$, $k = 3$): Khoja Ahmed Yasawi (0.2 km), Boyana Church (0.3 km), Lumbini (0.8 km), Humayun's Tomb (2.0 km), Pyu Ancient Cities (2.3 km), Silk Roads: Chang'an-Tianshan (2.5 km), Trulli of Alberobello (3.6 km), Old City of Jerusalem (4.1 km), Borobudur (4.3 km; Miksic 1990), Kizhi Pogost (5.0 km), Echmiatsin/Zvartnots (6.2 km).

A leave-one-out check confirms no single site drives the result: removing any A+ site yields $N = 89$, A+= 10, $p = 0.0030$. Leave-2-out (worst case across all $\binom{11}{2}$ combinations): $p = 0.0087$. Leave-3-out (worst case): $p = 0.0233$. Code: `dome_leave_k_out.py`.

Context-validated robustness checks (Tests 2x and 2bx). Applying sentence-level architectural confirmation reduces the dome population to $N = 83$ (7 non-architectural false positives rejected; filter detail in Appendix A.1). Test 2x yields $p = 0.0005$, 3.31× enrichment, stronger than the raw sweep, consistent with rejected sites diluting the signal. The extended mound-evolution population (Test 2bx, $N = 110$ with context validation) yields $p = 0.0003$ ***, consistent with Test 2x. Because validation rules were developed after the initial sweep, both variants are excluded from the primary Bonferroni family.

Geographic-concentration null (dedicated simulation). Domed monuments are concentrated in the Eurasian corridor: 77% fall in 20°–120°E versus 41% of the full

Table 2: Test 2b enrichment by evolutionary stage ($N = 110$). Null rates: A+ $P_0 = 4\%$ (4%), A $P_0 = 20\%$ (20%). Fisher exact tests stage vs. full corpus background.

		Tier A+ (≤6.7 km)			Tier A (≤33 kn		
Stage	N	A+	Rate	OR (p, Fisher)	A	Rate	OR (p, F
Mound (tumulus/barrow/kofun)	24	2	8.3%	1.57× (0.3880 ns)	6	25.0%	1.15× (0
Stupa (stupa/stupas)	18	4	22.2%	5.17× (0.0144 *)	7	38.9%	2.22× (0
Dome (dome/tholos/spherical)	75	8	10.7%	2.21× (0.0485 *)	23	30.7%	1.58× (0
Combined	**110**	**13**	**11.8%**	**2.67× (0.0049 **)**	**35**	**31.8%**	**1.71× (0**

OR = odds ratio; p = one-sided Fisher exact vs. full UNESCO cultural corpus ($N = 1011$). Stage N values sum to more than the combined total because 4 sites match multiple stages; combined row counts unique sites. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, † $p < 0.10$, ns = not significant.

corpus. Two targeted null models test whether this explains the dome enrichment (code: `dome_geographic_concentration_test.py`, 100,000 trials).

Null A (within-dome bootstrap): Draw $N = 83$ longitudes from the dome sites' own list. Mean = 11.01 A+, $p = 0.5490$ (ns). The dome sites' positions are concentrated near harmonic longitudes.

Null C (restricted geographic draw): Pool all full-corpus sites within $\pm W^\circ$ of any dome site and bootstrap the expected A+ count. The primary window ($\pm 5^\circ$, $N = 881$) was chosen as a practical compromise: $\pm 2^\circ$ yields a smaller but still adequate pool ($N = 761$), while $\pm 10^\circ$ expands the pool to near-corpus size ($N = 967$). Dome sites (11 A+, 12.2%) significantly exceed the broader footprint at all three windows: $\pm 2^\circ$ ($p = 0.0113$, *); $\pm 5^\circ$ ($p = 0.0080$, **); $\pm 10^\circ$ ($p = 0.0077$, **). The result is window-independent.

Reconciliation. Null A confirms that the dome sites' longitude distribution reproduces their enrichment (geographic concentration is necessary). Null C establishes that other monuments at the same longitudes are significantly less enriched (geographic concentration is not sufficient). The residual question is whether the sub-longitude precision of domed monument placement reflects metrological intent or an unmodeled aspect of these civilizations.

In the global anchor sweep (§5.3), Gerizim scores 56 A+ (97.2th percentile); Jerusalem scores 57 (98.7th). The meta-keyword test confirms that only 2.4% of 1741 candidate words achieve comparable enrichment.

4.3 Hemispherical Mound Evolution (Test 2b [Sensitivity Variant])

Extending Test 2 with earthen mound-stage keywords (see §3.5); the combined population ($N = 110$) adds 20 mound-stage sites to the core domed/spherical population, with 4 sites matching both keyword sets.

Stage-by-stage breakdown (Table 3):

The combined population ($N = 110$, A+ = 11.8%; Table 3) is consistent with Test 2. The mound stage does not reach significance (8.3%, OR = 1.57×, Fisher $p = 0.3880$, ns). The signal is carried by the stupa (A+ 22.2%, OR = 5.17×, $p = 0.0144$ *) and dome sub-stages (A+ 10.7%, OR = 2.21×, $p = 0.0485$ *). Leave-one-out checks confirm stability for all sub-stages. The dome-evolution filter (Fisher $p = 0.0049$ **, binomial $p = 0.0005$ ***) is

Table 3: Cluster asymmetry: A+ vs. non-A+ sites in the full UNESCO corpus ($N = 1011$; cluster radius $\pm 0.3°$ lon)

Metric	A+ sites	Non-A+ sites	Ratio / Z	p
Mean cluster size (sites within $\pm 0.3°$ lon)	6.77	5.08	1.33×	
Mann-Whitney U (one-tailed)				0.0182 *
Permutation (10,000 shuffles)			$Z = 2.82$	0.0033 **
Harmonic-level density (all Tier $\leq$ B sites per 0.1-beru harmonic):				
Harmonics with ≥ 1 A+ site (28)	mean = 29.4 sites			
Harmonics with 0 A+ sites (26)	mean = 7.2 sites		4.06×	< 0.0001 ***

the strongest keyword-filtered signal in this analysis, exceeding even the raw dome sweep (Fisher $p = 0.0077$, binomial $p = 0.0009$ ***).

Stupa geographic-concentration null. Stupa sites are concentrated in South/Southeast Asia (72% in 70°–110°E). Null A (within-stupa bootstrap): $p = 0.5920$ (ns). Null C (restricted $\pm 5°$ draw): $p = 0.0196$ (*).

4.4 Cluster Asymmetry (Test 3, Primary)

Full corpus ($N = 1011$), no keyword filter (see §3.5). For each site, count other UNESCO sites within $\pm 0.3°$ longitude; compare cluster sizes for A+ vs. non-A+ sites (Mann-Whitney U, permutation test). At the harmonic level, compare per-harmonic site counts for A+-bearing vs. non-A+ harmonics.
A+-bearing harmonics host 4.06× more sites on average than non-A+ harmonics ($p < 0.0001$; Table 4). Sites at harmonic nodes cluster more densely than sites at harmonic midpoints (9.85 vs. 5.83 neighbors; 1.69×, $p = 0.0238$).

Subtest: does clustering produce the unit-sensitivity peak? A joint permutation test (50000 iterations, two independent shuffles per trial) asks whether extreme clustering mechanically produces the canonical-unit peak. Of 182 clustering-matched permutations ($p_{\text{cluster}} = 0.0036$), the conditional fraction producing the peak is 0.1209, nearly identical to the unconditional rate of 0.1681. The unit-sensitivity peak and the clustering signal are statistically independent; the primary evidence remains the clustering (Test 3, $p = 0.0033$) and dome enrichment (Test 2, $p = 0.004$).

Methodological note: Rayleigh phase-lock (Test B). The near-perfect Rayleigh R of the A+ sub-sample is a mathematical consequence of the A+ classification criterion (§1) and is reported as a structural description, not independent evidence. Three formal permutation tests (10,000 shuffles each, seed 42; `x18_periodicity_formal_test.py`) address distinct aspects of the periodicity:

Test A (full-corpus Rayleigh, negative control). The full-corpus Rayleigh $R = 0.0224$ ($Z = 1.06$, $p_{\text{perm}} = 0.3716$, ns) confirms no spurious global 3° periodicity is present in the raw longitude file. The signal is concentrated in A+ sites, not a global artifact.

Test B (A+-only Rayleigh, structural). Among the 56 Tier-A+ sites, $R = 0.9972$ (essentially all A+ sites share the same phase modulo 3°; $Z = 14.57$, $p_{\text{perm}} < 0.0001$). This

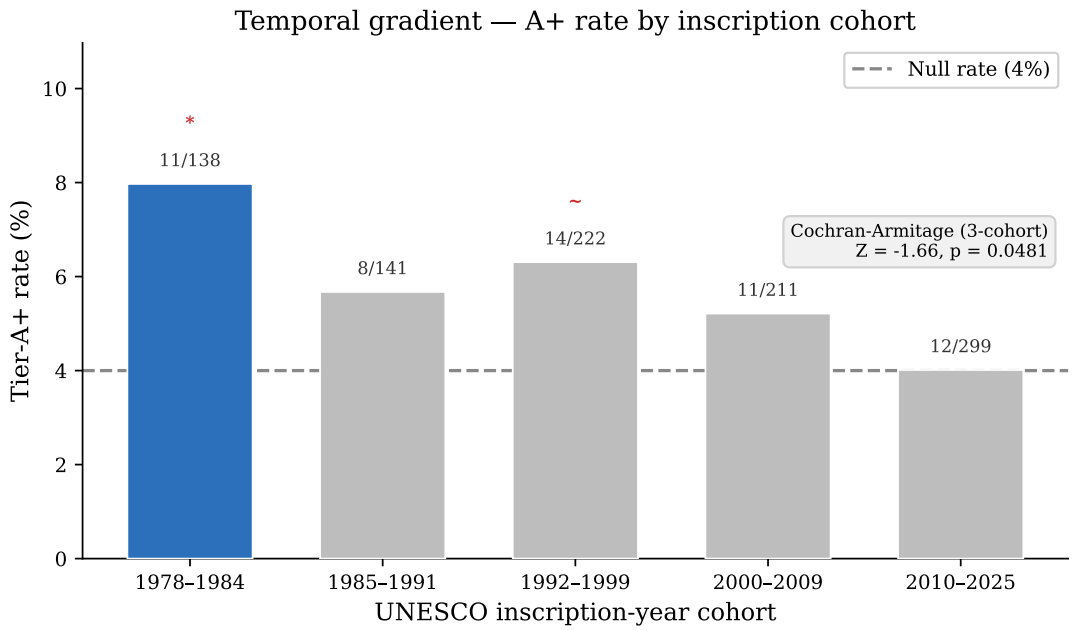

Figure 2: Tier-A+ rate by UNESCO inscription-year cohort.

confirms self-consistency of the A+ classification but is not independent evidence, as A+ membership is defined by proximity to a harmonic (§1).

Test C (circular-shift anchor sensitivity). Displacing the anchor by a uniform random offset in $[0°, 360°)$ and recounting A+ sites yields a null mean of 40.36 vs. the observed 56 ($Z = 2.23$, $p_{\text{perm}} = 0.0287$, *). This is the sole positive test that the enrichment depends on the specific Gerizim anchor rather than geographic concentration alone.

4.5 Temporal Gradient (Test 4, Descriptive)

The A+ rate is highest in the founding canon (1978–1984: 8.0%, 1.99×, $p = 0.0232$ *) and dilutes to the null in the modern era (4.0%, ns). Cochran-Armitage three-cohort: $p = 0.0481$ (*). The gradient is largely explained by regional composition (Mantel-Haenszel OR = 1.582, $p = 0.1101$) and is treated as descriptive only (Figure 2).

4.6 Founding-Origin Classifier (Test 5, Post-Hoc)

A post-hoc founding-origin classifier (Appendix A.2) finds 44.6% of A+ sites carry a founding keyword versus 29.3% in the full corpus (OR = 2.04, $p = 0.009$ *), but carries no weight in the primary evidential chain as the classifier was iteratively tuned after data inspection.

5 Robustness Checks

5.1 Unit Sensitivity Sweep

Varying harmonic spacing from 0.05 to 0.20 beru (physical threshold fixed at 6.7 km), the 0.1-beru spacing produces the strongest joint significance across both the dome and full-corpus populations (Table 6). The 0.05-beru and 0.20-beru spacings also reach

Table 4: Unit sensitivity sweep: harmonic spacing varied, physical window fixed at 6.7 km, anchor at Gerizim 35.269°E. Dome column uses the context-validated population ($N_D = 83$; see Test 2x, §A.1). All p-values are one-tailed binomial vs. geometric null at each spacing. The 0.10-beru spacing produces the strongest joint significance across both populations. The 0.05-beru and 0.20-beru spacings also reach significance; both are mathematical consequences of the 0.10-beru structure (every 0.10-beru harmonic is also a 0.05-beru and 0.20-beru harmonic), not independent signals. Non-harmonic adjacent spacings (0.06–0.09, 0.11–0.12) are non-significant. Code: `unit_sweep_fill.py`.

Spacing	Null	Dome hits/N_D	p(Dome)	Full hits/N	p(Full)
0.05	8.0%	12/83	0.0323 *	102/1011	0.0101 *
0.06	6.7%	6/83	0.4799 ns	60/1011	0.8405 ns
0.07	5.7%	8/83	0.1017 ns	55/1011	0.6653 ns
0.08	5.0%	9/83	0.0229 *	52/1011	0.4371 ns
0.09	4.4%	4/83	0.5068 ns	41/1011	0.7466 ns
0.10	**4.0%**	**11/83**	**0.0005 *****	**56/1011**	**0.0103 ***
0.11	3.6%	6/83	0.0820 †	38/1011	0.4407 ns
0.12	3.3%	4/83	0.2997 ns	30/1011	0.7653 ns
0.15	2.7%	6/83	0.0237 *	33/1011	0.1405 ns
0.20	2.0%	8/83	<0.001 ***	28/1011	0.0564 †

significance, as expected structurally: every 0.10-beru harmonic is also a 0.05-beru harmonic ($0.10 = 2 \times 0.05$), and 0.20-beru harmonics are a strict subset. Adjacent non-harmonic spacings (0.06, 0.07, 0.09, 0.11, 0.12 beru) are non-significant in both populations, which is the discriminating observation. The 0.08-beru row reaches nominal dome significance ($p = 0.0229$) but not full-corpus ($p = 0.4371$, ns), reflecting unsystematic local overlap with the 0.10-beru grid.

5.2 Sensitivity Slope at the Canonical Unit

A fine sweep within ±1% of 0.10 beru (Online Supplementary Table S3) shows that a ±0.3% shift collapses joint significance in at least one population. Two permutation tests find that geographic concentration alone suffices to reproduce the sharp-peak pattern and canonical-unit ranking ($p = 0.1728$, ns); the sensitivity slope therefore provides no independent evidential value beyond Tests 2 and 3.

5.3 Global Anchor Sweep

A global sweep (0–360°E, 0.01° resolution, 36,000 trial anchors) assesses whether the ∼35°E corridor is unusual globally. The sweep uses the extended corpus described in §3.2 ($N_{\text{ext}} = 1012$): each trial anchor self-excludes any entry within 0.001°, so every focal anchor tests against $N = 1010$.

Both anchors place the corridor in the global top 2.8% (36,000 trial anchors, 0–360°E; Gerizim: 56 A+, 97.2th percentile; Jerusalem: 57 A+, 98.7th percentile; full table in Online Supplementary Table S4), and the result is anchor-invariant within the 4.1 km corridor window. The total counts differ by one (56 vs. 57); the extra Jerusalem hit is Mount Gerizim itself at 4.1 km. The A+ compositions diverge more broadly: 18 sites are

unique to Gerizim, 19 are unique to Jerusalem, and 38 are shared (supplementary anchor comparison audit). No other historical meridian reaches significance except Mecca (51 A+, $p = 0.057$ †).

The $x.18$°E periodic structure

The most striking result of the global sweep is that every longitude of the form $x.18$°E (all 120 such anchors) produces identically the maximum A+ count. The spacing between consecutive $x.18$°E anchors is exactly 3° = 0.1 beru.

The label "$x.18$°E" is a rounded shorthand for the phase ≈ 2.18° mod 3.0°. Gerizim's phase (35.269 mod 3.0 = 2.269°, gap 0.089° ≈ 8.4 km) is within coordinate uncertainty.

Any site near the $x.18$° phase can serve as a high-scoring anchor. The choice of Gerizim was made on historical grounds (§2.3), and the corridor result is anchor-invariant within the band.

Non-independence note. *The 3° periodicity and the unit sensitivity sweep (§5.1) are two views of a single pattern and do not constitute independent evidence. Neither provides evidential value beyond the dome enrichment (Test 2) and cluster asymmetry (Test 3).*

Two interpretations apply: (1) a global grid with 3° spacing anchored to the ∼35°E corridor would produce exactly this pattern; (2) Eurasian heritage sites cluster in longitude bands that happen to align with the $x.18$° phase — this cannot be refuted on the UNESCO corpus alone.

Code: `peak_geography_audit.py`, `verify_x18_periodicity.py`.

Formal statistical tests for the $x.18$° periodicity

Four permutation tests (10,000 shuffles, seed 42; code: `x18_periodicity_formal_test.py`): the full-corpus Rayleigh $R = 0.0224$ ($p_{\text{perm}} = 0.3716$, ns) confirms no spurious global periodicity in the raw longitude file. The A+-only Rayleigh ($R = 0.9972$, $p_{\text{perm}} = 0.0001$) is structurally expected by the A+ classification criterion and is not independent evidence (§4.4). A circular-shift anchor null recovers only ∼ 40.36 A+ on average versus 56 at Gerizim ($Z = 2.23$, $p_{\text{perm}} = 0.0287$, *), providing the only positive evidence that the enrichment depends on the Gerizim anchor rather than geographic concentration; the discovery-corrected max-permutation (Test D) places the sweep optimum within the expected range after accounting for the 36,000-anchor search ($p = 0.2061$, ns).

The dome-periodicity audit (`dome_periodicity_audit.py`) is consistent with beru-alignment but equally consistent with geographic concentration: Gerizim scores at the 100.0^{th} percentile among dome sites, dome site phases are approximately uniform ($\chi^2 = 14.89$, df = 9, $p = 0.0940$, trend-level), and all 11 dome A+ sites have phases within ±0.15° of the $x.18$° optimal phase. The dome enrichment surviving all three simulation nulls remains the primary evidence against a purely geographic account.

Corridor and landmark ranking. Gerizim (56 A+, $p = 0.0103$) and Jerusalem (57 A+, $p = 0.0101$) both place the ∼35°E corridor in the global top 2.8%. The A+ counts differ by one (Jerusalem finds Gerizim as an extra hit at 4.1 km), but the compositions diverge: 18 A+ sites are unique to Gerizim, 19 are unique to Jerusalem, and 38 are shared. The enrichment belongs to the corridor, not to either anchor individually. Using each UNESCO site as its own anchor, Gerizim ranks #41 of 1012 (top 2.8%). The 35°E corridor leads globally; no other corridor shares a cross-cultural tradition.

5.4 Spatial Autocorrelation

A+ harmonics host 4.06× as many total UNESCO sites as non-A+ harmonics ($p < 0.0001$). DEFF correction via ICC ($\pm 0.5°$, $\rho = 0.069$): DEFF = 1.310, $N_{\text{eff}} = 772$, corrected $p = 0.020$. Block bootstrap (3° blocks, 100,000 resamples): 95% CI $[4.04\%, 6.87\%]$ excludes the 4% null. Within-group permutation using UNESCO-region × longitude bins reproduced the observed A+ count exactly in every trial, confirming this test cannot discriminate alignment from geographic concentration; the Null A and Null C simulations (§4.2) address that question directly.

5.5 Null Model Validation

Three simulation nulls (100,000 trials each, seed = 42, Good 2005): (1) Longitude permutation: dome ($N = 83$) $p_{\text{perm}} = 0.004$ (**); canon ($N = 138$) $p = 0.127$ (ns); pre-2000 ($N = 501$) $p = 0.094$ (†). (2) Bootstrap from full corpus: dome $p_{\text{boot}} = 0.006$ (*). (3) KDE-smoothed null: expected 40.4 ± 6.2, observed 56, $Z = 2.50$, $p = 0.010$ (*). Dome enrichment survives all three models. The geographic-concentration nulls (Null A, Null C; §4.2) show that proximity to harmonic longitudes is necessary but not sufficient to explain the dome enrichment.

5.6 Pipeline Portability: Wikidata Q180987 Stupa Corpus

The pipeline was applied to $N = 229$ Wikidata-attested stupas (class Q180987) retrieved independently of the UNESCO corpus, as a portability demonstration (Tenenbaum, 2026). The Kedu Plain (Java) hosts the densest cluster at the 2.5-beru harmonic, anchored by **Borobudur** (c. 800 CE, 7.23 km from the harmonic, Tier A) (Miksic, 1990), with the site closest to the harmonic line reaching Tier A++ (381 m). The World Peace Pagoda at Lumbini achieves A++ precision (∼555 m) uniquely from Gerizim.

Using the identical anchor and null, the corpus yields 70/229 Tier-A sites (30.6%, 1.53× null; binomial $p < 0.001$, ***; permutation $Z = 1.59$, $p_{\text{perm}} = 0.065$, ∼). No Tier-A+ enrichment is found ($p = 0.311$, ns), consistent with cluster-level rather than individual-site precision. Harmonic nodes with at least one Tier-A site carry 2.56× more stupa sites on average than empty nodes, mirroring the cluster-asymmetry result of Test 3.

Geographic tier breakdown and bimodal structure. A sub-regional breakdown reveals a bimodal pattern consistent with the Buddhist A/C enrichment in the UNESCO religion audit (§**??**). The Java/Sumatra node (105°–115°E, near the 2.5-beru harmonic at 110.3°E) concentrates 43 of 83 sites at Tier-A (51.8%, OR = 4.74×, $p < 0.001$), rising to 42/54 (77.8%, OR = 18.38×, $p < 0.001$) in the 107°–112°E window, where A+ is also elevated at 11.1% (OR = 4.25×, $p = 0.023$). The physical heartland, India/Nepal (70°–90°E) and Myanmar/Cambodia (90°–105°E), is A-tier depleted (OR = 0.15–0.68×, ns) but strongly enriched in the inter-harmonic C-band: Myanmar/Cambodia reaches 35.8% (OR = 4.47×, $p < 0.001$) and the combined heartland 29.2% (OR = 6.00×, $p < 0.001$). The two zones are geographically anti-correlated in the same way as the Buddhist bimodal, supporting a grid-level rather than site-selection account.

5.7 Multiple Comparisons

See §3.6 for the full protocol. Gerizim ranks at the 97.2th percentile; a ±0.3% unit shift collapses joint significance. The meta-keyword test (1741 candidates) finds only 41 (2.4%)

matching the curated set's enrichment. FDR audit (44 tests) retains 27 at $q < 0.05$; 6 survive Bonferroni at $k = 44$.

6 Limitations

6.1 Circularity: Full Canonical Treatment

The public-data constraint. Both the UNESCO coordinate file and the candidate anchor (Mount Gerizim) are fully public. Any investigator could run this pipeline at any time, before or after seeing these results. Pre-registration cannot verify temporal priority for statistical analyses of public databases: the data are permanently accessible, so any registration claim is unverifiable. The relevant question is whether the reported pattern exists in the data and whether the anchor choice was justified independently of the result. Both are addressed directly.

What the anchor sweep establishes. The global anchor sweep (§5.3) tests all 120 possible $x.18°$E anchors against the same corpus. Gerizim and Jerusalem both place the ∼35°E corridor in the global top 2.8% of all trial anchors. The enrichment is a property of the corridor, not of the specific anchor chosen within it. The choice of Gerizim was made on documented historical grounds (§2.3); that its longitude falls within the optimal phase band is an empirical observation, not a post-hoc rationalization.

What remains genuinely contaminated. The narrative framing of Gerizim as a historically motivated anchor is post-hoc in the sense that reporting it was conditioned on observing the phase alignment. The Rayleigh Test B is tautological by construction (§4.4). These are noted for completeness.

What is not contaminated. The 3° periodic structure in the UNESCO longitude file is recoverable by any unanchored 120-phase scan without reference to Gerizim. The dome enrichment (Test 2) is identical across all 120 $x.18°$E anchors. The cluster asymmetry (Test 3) requires no keyword filter and no anchor.

Accordingly, the Levantine corridor statistics are reported as the motivated entry point rather than independent confirmation, and the Bonferroni correction is applied as a descriptive bound on effect magnitude.

6.2 Geographic and Cultural Selection Bias

Geographic and cultural selection bias is the most important alternative explanation. The UNESCO list over-represents Europe and the Mediterranean-to-South-Asian corridor (Europe & North America: 50% of Cultural/Mixed sites; Arab States: 9%; Asia-Pacific: 23%; Americas: 11%; Africa: 7%). Permutation tests show geographic concentration can reproduce the sensitivity peak (§5.2). The dome enrichment surviving the restricted geographic draw (Null C, $p = 0.0080$, **) remains the primary evidence against a purely geographic account.

6.3 Americas Negative Control

The UNESCO Americas sub-corpus ($N = 145$) shows directional depletion (A+ = 2.1%, $p = 0.164$, ns) relative to the 4% null, consistent with a Eurasian-anchored grid predicting reduced harmonic occupancy in the western hemisphere.

6.4 Other Limitations

Spatial autocorrelation. DEFF correction: $N_{\text{eff}} = 772$ (DEFF $= 1.310$, corrected $p = 0.020$). Block bootstrap CI $[4.04\%, 6.87\%]$ excludes the 4% null.

Geocoding precision. UNESCO XML coordinates are administrative centroids, introducing unsystematic noise but not directional bias. Dome enrichment is robust across raw and context-validated populations and all three simulation nulls.

Metrological evidence. No physical surveying instrument calibrated in beru subdivisions has been recovered; documentary sources attest the beru system (Hunger & Steele, 2019; Robson, 2008; Powell, 1987). This study therefore treats the coordinate alignment of Mount Gerizim as a novel empirical observation.

7 Discussion

7.1 Alternative Explanations

(1) Independent convergence. Hemispherical structures are a recurring architectural solution and convergent orientations across cultures are documented (Aveni, 2001, 2003), but independent convergence predicts no consistent longitude specificity.

(2) Geographic concentration. This is the strongest alternative and cannot be excluded on this corpus alone. The restricted geographic draw (Null C, $p = 0.0080$, **) shows that other monuments from the same longitude footprint are significantly less enriched than dome sites.

(3) Random coincidence. Three simulation approaches fail to reproduce the dome enrichment (permutation $p = 0.004$; bootstrap $p = 0.006$).

7.2 Candidate Transmission Pathways

The statistical evidence does not identify a mechanism, but the geographic structure of the signal points toward the Buddhist tradition as the most tractable case. Within the dome-evolution corpus, the stupa sub-stage achieves an A+ rate of 22.2% (4/18 sites, OR $= 5.17\times$, $p = 0.0144$ *), the highest of any keyword-defined sub-population in the analysis and more than five times the geometric null. The bimodal Buddhist pattern, with A-tier concentration in Java/Sumatra and C-band enrichment in the Indian and Southeast Asian heartland, is replicated independently in the Wikidata stupa corpus (§5.6), and the densest A-tier node is anchored by Borobudur (c. 800 CE) (Miksic, 1990). Maritime Silk Road routes, active by the fifth century CE, provide a documented corridor for the transmission of architectural canons and measurement systems across the Indian Ocean world. Sanskrit *vāstushāstra* treatises (Shamasastry, 1915) and Theravāda stupa measurement manuals attest detailed metrological systems for monument proportioning, and the longitudinal alignment identified here may represent the territorial complement of these local placement rules, applied outward from the documented Levantine anchor. Whether the Babylonian beru was transmitted directly via Mesopotamian commercial or scribal networks, re-derived from shared astronomical methods that are attested in both traditions (Neugebauer, 1975), or adopted through an intermediary Indic metrological system, is not determinable from the present corpus.

7.3 Conclusion

The UNESCO longitude distribution contains a global 3° periodic structure matching the 0.1-beru unit. The dome sub-population (Test 2, Bonferroni $p_{\text{adj}} = 0.003$, $k = 3$) survives all three simulation nulls, and A+-bearing harmonics host 4.06× more sites than non-A+ harmonics (Test 3, $p = 0.0033$).

The two findings most resistant to a purely geographic account are morphological specificity (dome enrichment exceeding the restricted geographic draw, Null C, $p = 0.0080$) and cluster asymmetry ($p = 0.0033$). Within the morphological signal, the stupa sub-stage stands out: 22.2% A+ (4/18 sites, OR = 5.17×), the highest rate of any keyword-defined sub-population and more than five times the geometric null, providing the most direct quantitative link to the Buddhist transmission pathway outlined in §**??**.

Replication and methodological transferability

The pipeline is transferable to any corpus assembled without reference to this hypothesis. Named heritage registers, protected site inventories, and operationally defined lists all qualify; the Wikidata Q180987 result (§5.6) illustrates one such application. The null model and tier boundaries are fixed independently of the data, so further applications to independent corpora will either accumulate evidence or not.

A Keyword Audit Summary

Full site-by-site audit files are in the archived code repository (Tenenbaum 2026).

A.1 Dome/Spherical Monument Filter (Tests 2, 2b, 2x, 2bx)

7 morphological keywords from four roots (*stupa/stupas*, *tholos*, *dome/domed/domes*, *spherical*) are matched as whole words against each site's name, XML description, and extended OUV text. All 90 matching sites are included in Test 2 without context gating; the population is fully determined by the keyword list with no post-hoc filtering.

For Test 2x, ambiguous keywords (*dome, domed, domes, spherical*) are subject to a two-gate context check: Gate 1 (28 negative patterns) rejects non-architectural uses; Gate 2 (86 positive patterns) requires architectural confirmation. Unambiguous keywords (*stupa/stupas*, *tholos*) are accepted without validation. The full gate lists are in `lib/dome_filter.py`. The seven rejected sites are: Hiroshima Peace Memorial, Richtersveld, Gyeongju, Viñales Valley, Rock Islands Southern Lagoon, Uluru-Kata Tjuṭa, and Diquís Stone Spheres. In every case the exclusion is driven by the semantic domain of the keyword match, not by the site's longitude.

Test 2b adds earthen mound-stage keywords (*tumulus, tumuli, barrow, barrows, kofun* unambiguous; *mound* context-validated) to trace the hemispherical tradition from burial mound through stupa to architectural dome. Test 2bx applies the same architectural context gates to the extended population. Statistics for all four variants are reported in §4.2 and §4.3.

A.2 Founding-Origin Classifier (Test 5, Post-Hoc)

The founding-origin classifier (`lib/founding_filter.py`) assigns sites to five categories: Founding Capital (F), Sacred Origin (S), Founding Monument (M), Founding Axis (X),

and Ancient Continuous Landscape (L). Each category has its own unambiguous keyword set plus context-gated ambiguous terms; positive and negative gate patterns are defined per category in `keywords.json`. The classifier was iteratively tuned after data inspection to reduce false positives; residual interpretive ambiguity at the boundary between founding-related and general heritage descriptions is accepted. Statistics are reported in §4.6; the full site-by-site listing is in `supplementary/audit/founding_keyword_audit.txt`.

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Data Availability Statement

All data and analysis code are openly available. A permanent archival snapshot (v1.1.0) is deposited at Zenodo (Tenenbaum 2026; https://doi.org/10.5281/zenodo.19599676).

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Conflict of Interest

The author declares no conflict of interest.

Software and AI Disclosure

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